

CWA SCIENCE CLASSES

Concept With Abhishek | Class 9 — Physics

Chapter 1: Motion

1. Distance & Displacement: Distance is the total path length travelled by an object; displacement is the shortest straight-line distance between initial and final position along with direction.
2. Speed = Distance / Time (scalar quantity). Velocity = Displacement / Time (vector quantity).
3. Uniform Motion: Equal distances in equal intervals of time. Non-uniform motion: unequal distances in equal intervals of time.
4. Acceleration: Rate of change of velocity. $a = (v - u) / t$.
5. Equations of Motion: $v = u + at$, $s = ut + \frac{1}{2} at^2$, $v^2 = u^2 + 2as$.
6. Graphs: Distance-time graph slope = speed. Velocity-time graph slope = acceleration; area under v-t graph = distance covered.
7. Circular Motion: Motion along a circular path at constant speed is accelerated motion because direction keeps changing.
8. Important Numerical Tip: Always convert km/h to m/s by multiplying with $\frac{5}{18}$ before solving numericals.